UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Conversation 1

Welcome & ~~Lecture 1~~

Reference: Chapter 1
(Sections 1.1 – 1.2)

What this Course is About

This course is about signals and signal-processing systems

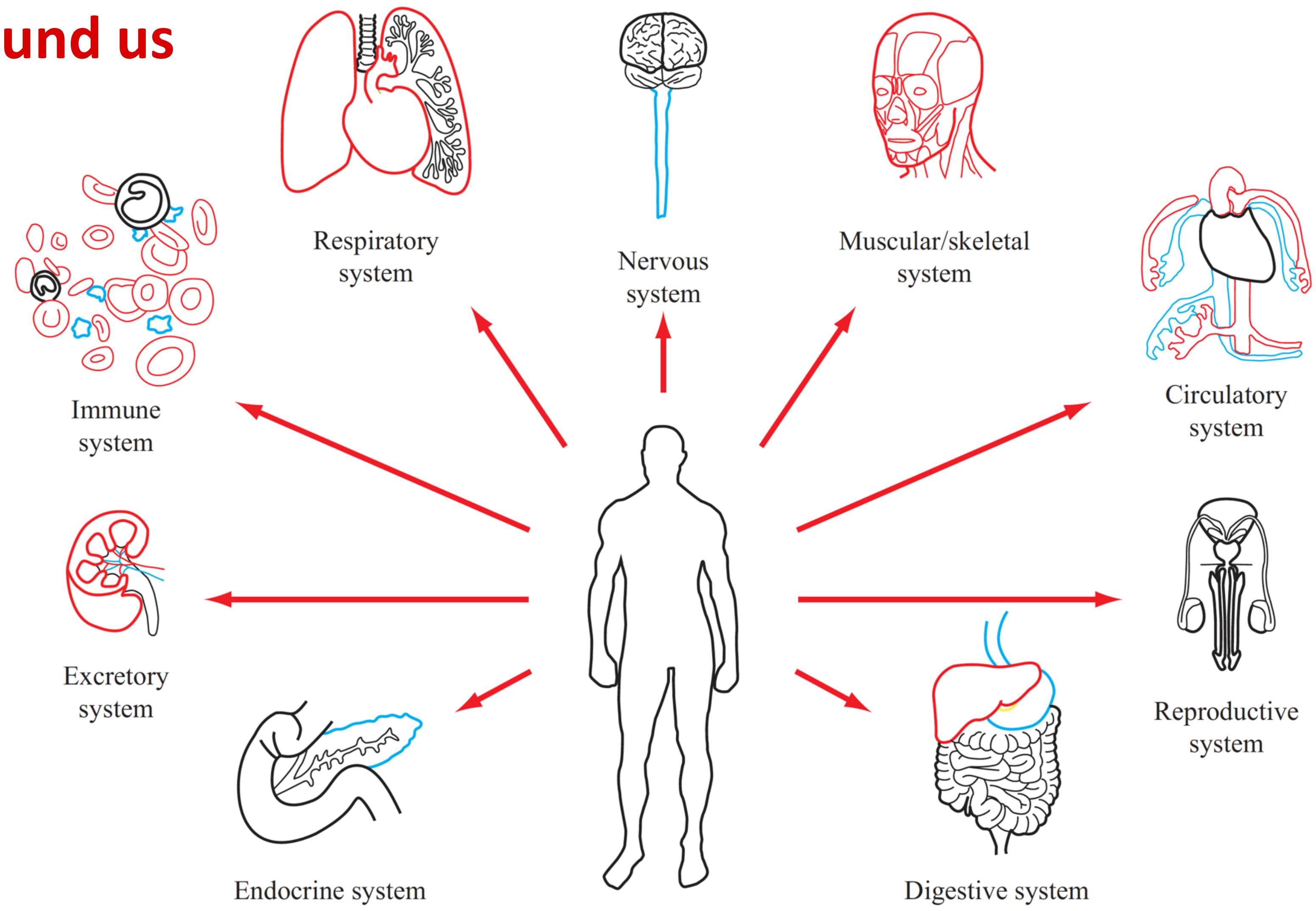
Definitions:

- **Signal:** A quantity (e.g., a voltage) that varies with time, space, or another independent variable
- **System:** An entity (e.g., a circuit) that can transform an input signal x(t) to a new signal y(t)

Signals and Systems are Everywhere!

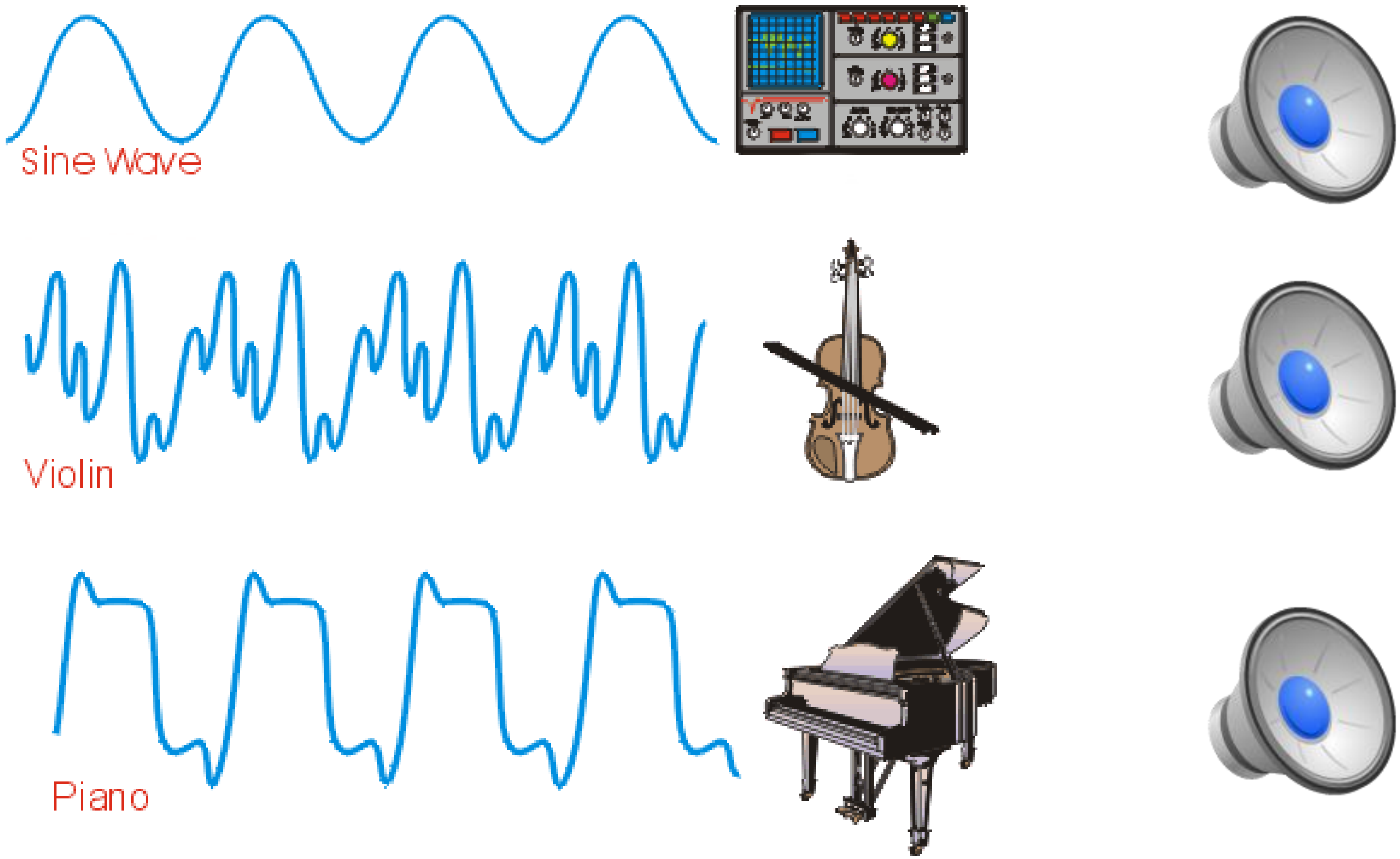
Signals and Systems are all around us

- Music
- Cell phones
- Television
- Computers
- Biological systems
- Medical Electronics
- Mining, Oil & Gas exploration
- Control systems
- Electric Power Grid
- GPS for navigation
- Military Systems …



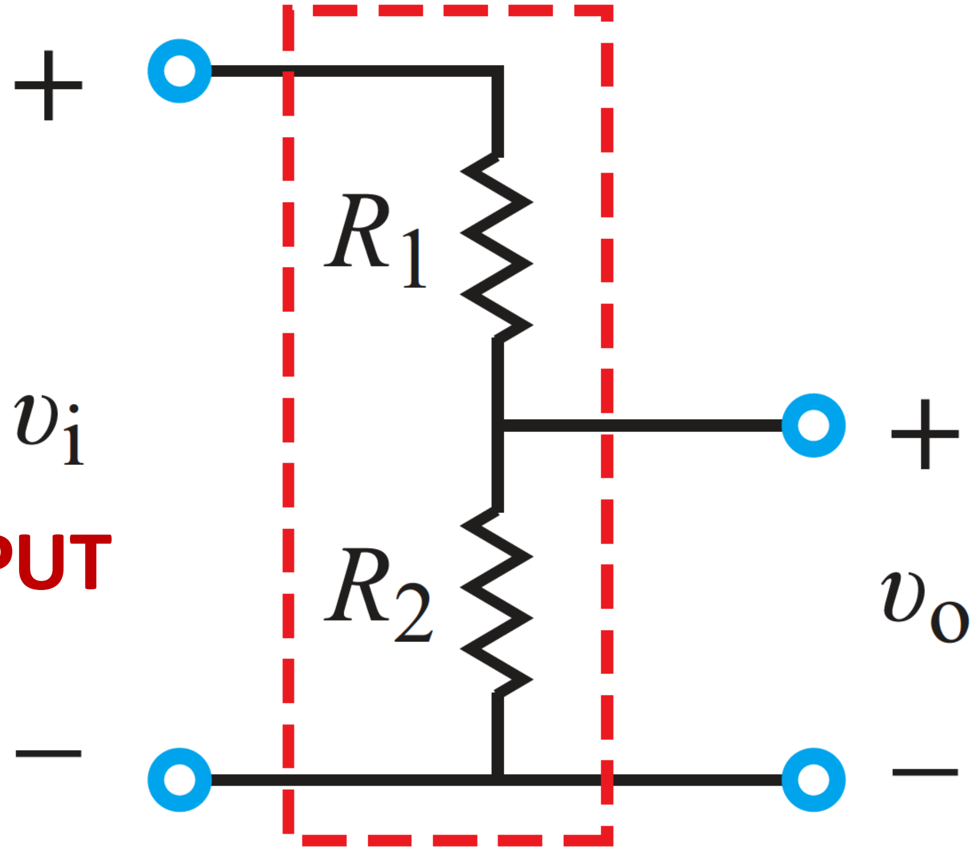
Analyzing signals and designing systems are core engineering skills

Example: Audio Signals



method-behind-the-music.com

Example: A Simple System



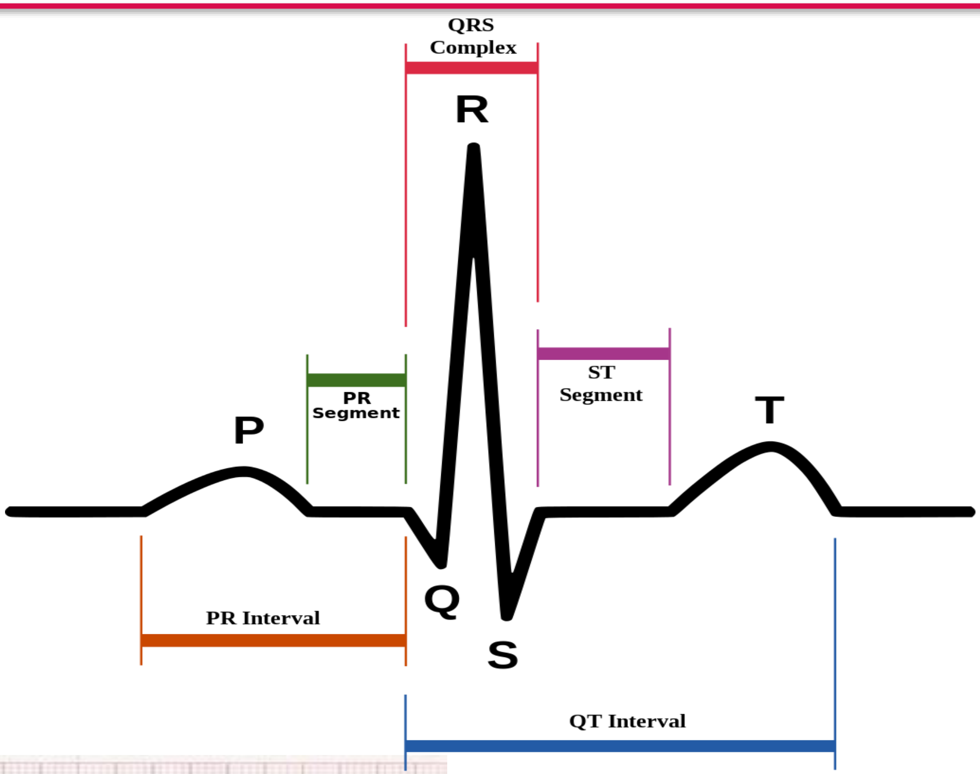
INPUT

$$v_o = \frac{R_2}{R_1 + R_2} v_i$$

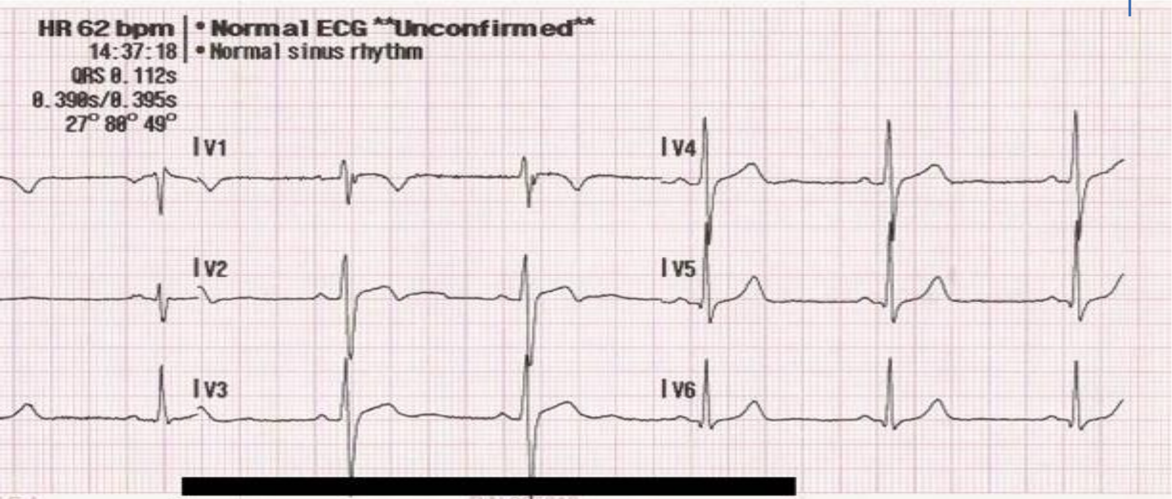
OUTPUT

VOLTAGE DIVIDER
(SYSTEM)

Example: Heart Signal (ECG)

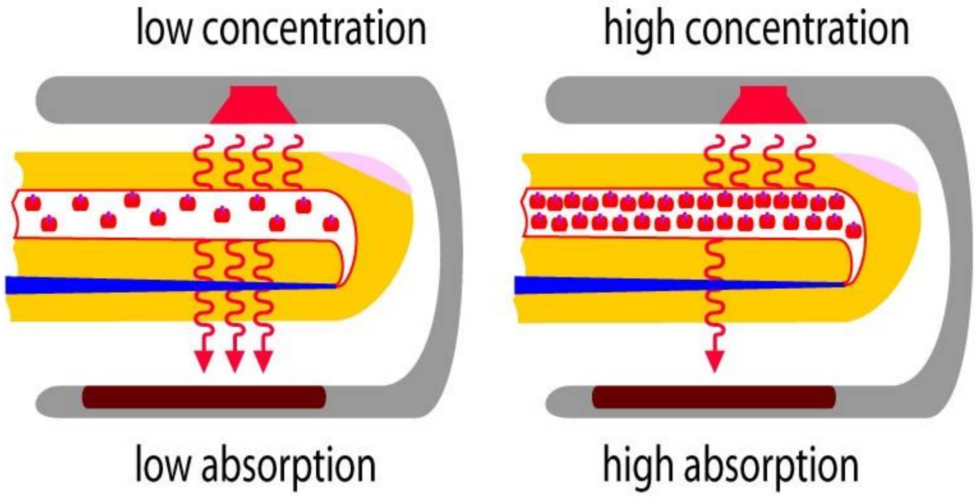


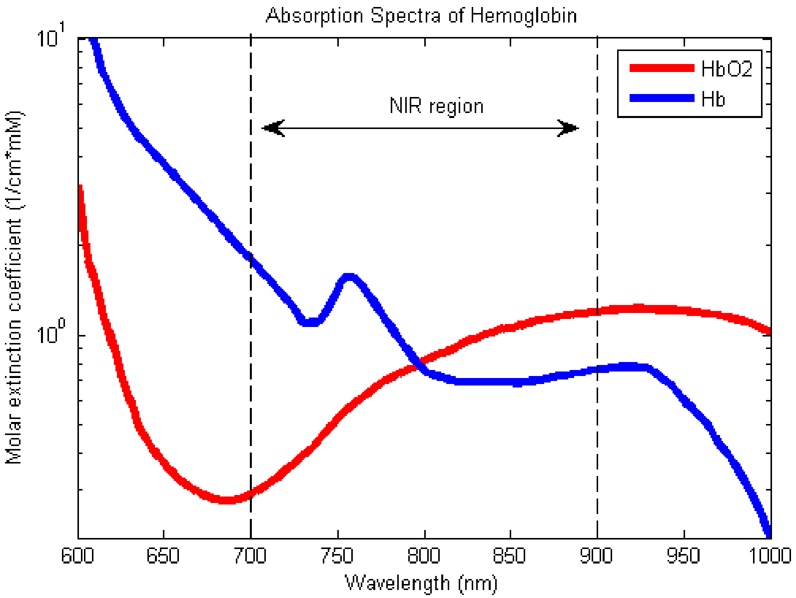
Irregularities in the shapes and durations of each part of these pulses indicate different health conditions



Heart monitor

Example: Pulse Oximeter





$s_{700}(t)$ $s_{900}(t)$

Signals From Optical Sensors

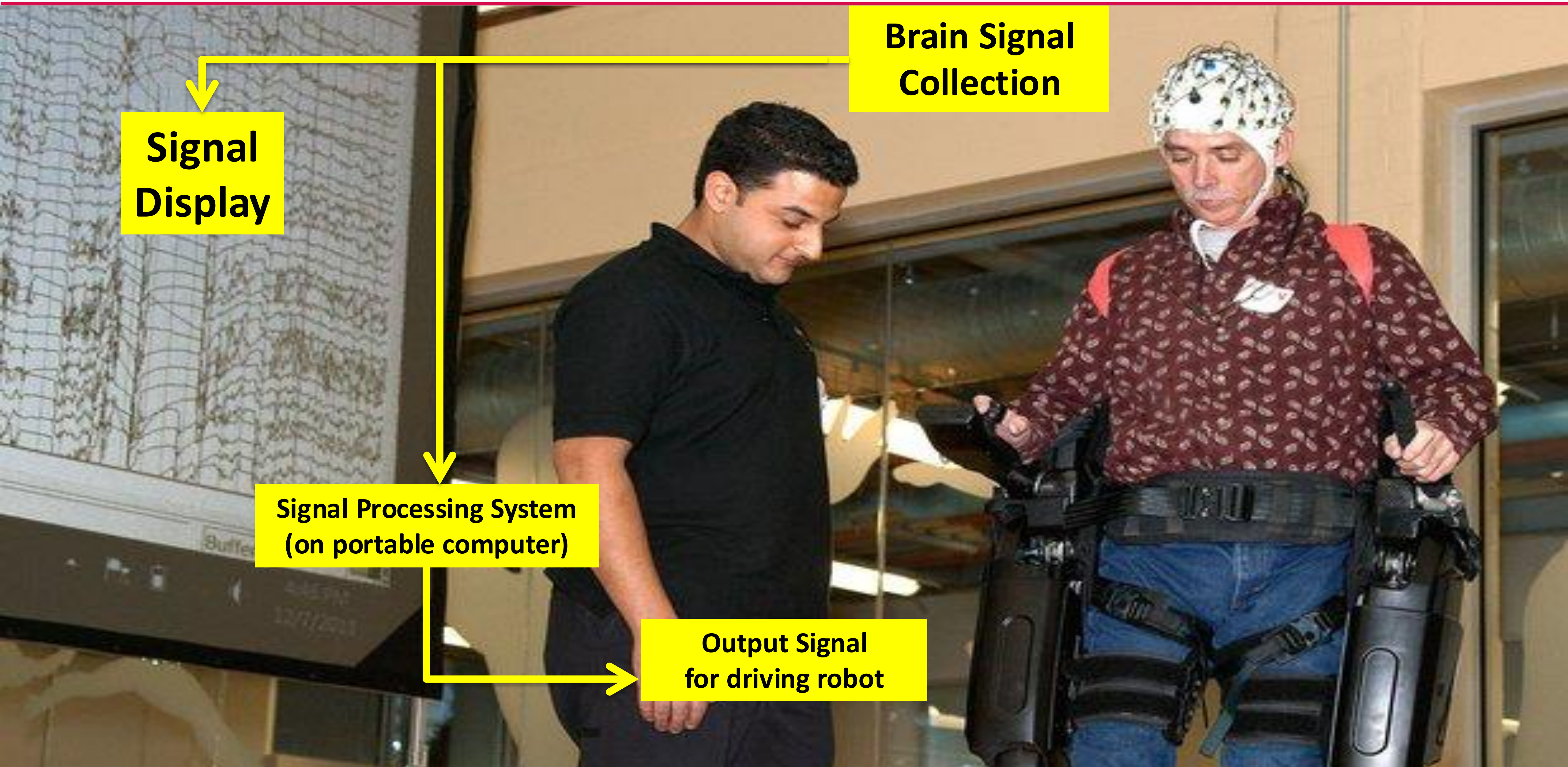
$s_{700}(t), s_{900}(t)$

System

Blood Oxygen Saturation Readout

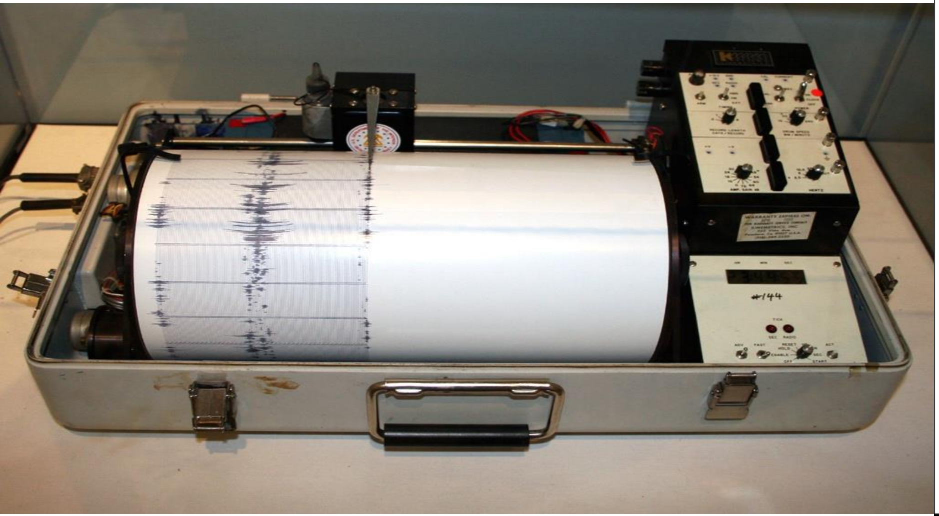
SpO_2

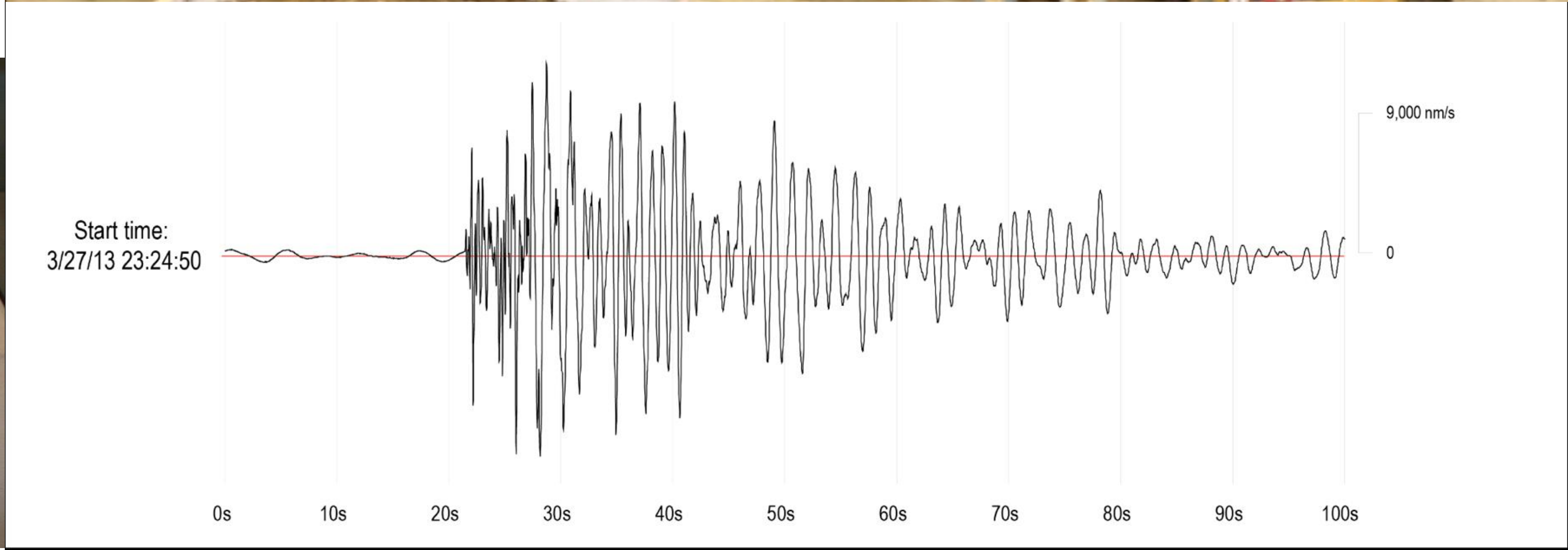
Example: Brain Activity Signals (EEG)



Mining Industry Example

- Open-pit mines set off explosions to break away rocks.
- The explosive jolt is an input signal to the ground (the system)
- A seismometer captures the resulting ground vibrations as the (output) signal.
- The signal patterns provide clues about the properties of the underground rocks





ECE 3337 Catalog Description

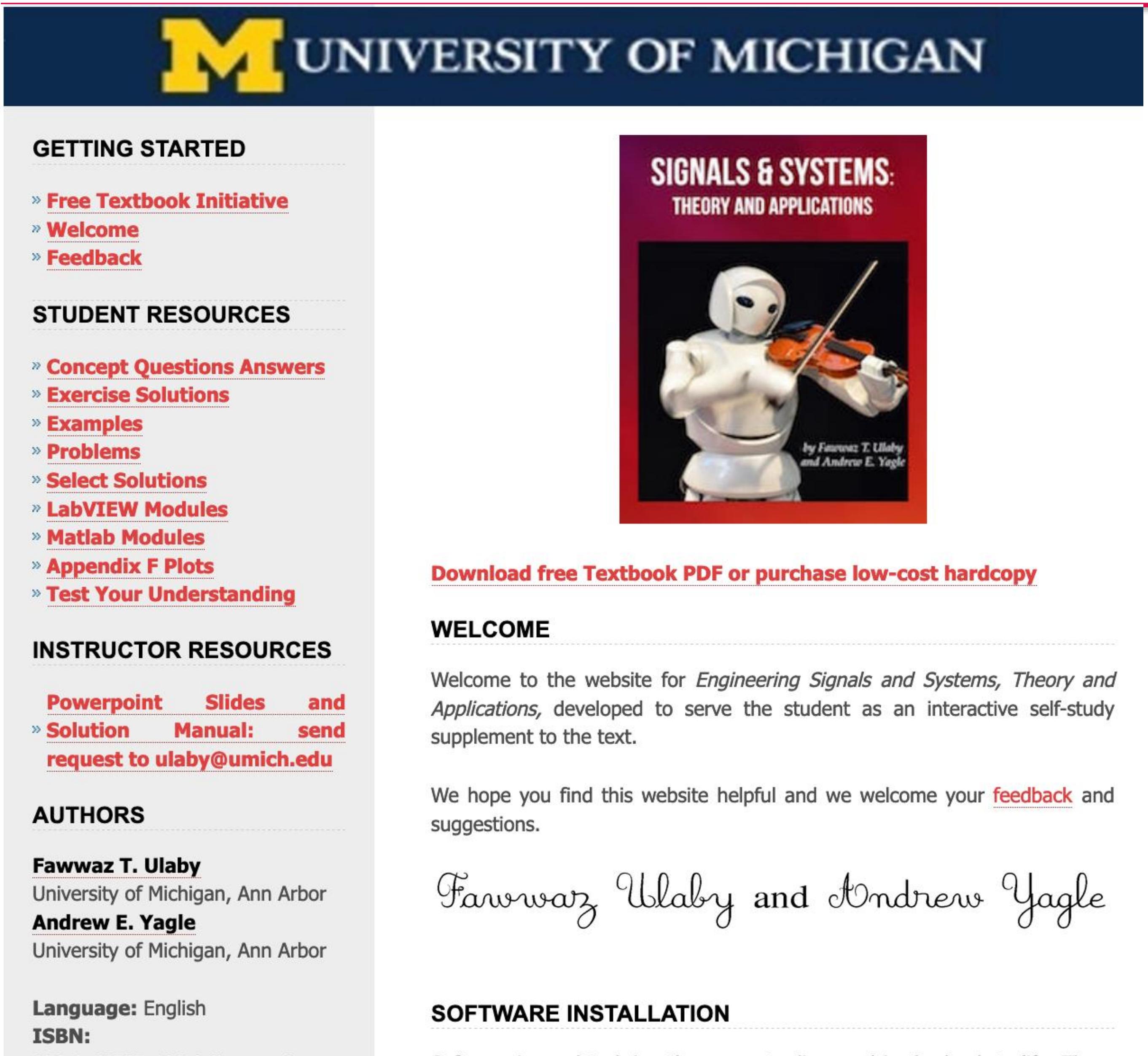
- Time and frequency domain techniques for signals and systems analysis.
- Engineering applications of the convolution integral, Fourier series and transforms, and Laplace transforms.

Course Pre-requisites:

- MATH 3321 (Engineering Math)
- ECE 2202 (Circuit Analysis II)

Basically, we will use a lot of complex variables, integral calculus, and circuit theory with resistors, capacitors, inductors, and op-amps

Action Item #1: Download the Free Textbook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GETTING STARTED

» Free Textbook Initiative
» Welcome
» Feedback

STUDENT RESOURCES

» Concept Questions Answers
» Exercise Solutions
» Examples
» Problems
» Select Solutions
» LabVIEW Modules
» Matlab Modules
» Appendix F Plots
» Test Your Understanding

INSTRUCTOR RESOURCES

» Powerpoint Slides and Solution Manual: send request to ulaby@umich.edu

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Language: English
ISBN:

SIGNALS & SYSTEMS: THEORY AND APPLICATIONS

by Fawwaz T. Ulaby and Andrew E. Yagle

Download free Textbook PDF or purchase low-cost hardcopy

WELCOME

Welcome to the website for *Engineering Signals and Systems, Theory and Applications,* developed to serve the student as an interactive self-study supplement to the text.

We hope you find this website helpful and we welcome your feedback and suggestions.

Fawwaz Ulaby and Andrew Yagle

SOFTWARE INSTALLATION

- We will follow the textbook closely
  - Chapters 1 – 6
  - Study ahead required
  - First edition is adequate if you prefer hardcopy
- Textbook website provides supplemental materials

FREE DOWNLOAD:
https://ss2-2e.eecs.umich.edu/

Action Item #2: Obtain the Zybook

- **Purchase the website version of the textbook (Zybook)**
  - **Zybook Code:** UHECE3337ReddyFall2025

- **Why are we using the Zybook?**
  - **The Zybook provides all of the textbook content plus built-in examples, animations, solution details, and quizzes with help features**
  - **Its automatic grading feature will be used to give you course points for studying in advance for each lecture**

- **Important: The Zybook does <u>not</u> contain the end-of-chapter exercises.**

How You Study Matters a Lot

You are expected to study *in advance* for each class

The last slide of each lecture tells you what will be covered next

Study Method:

1. Read the assigned textbook sections in order to understand the basic concepts, and aim to memorize the central concepts
2. Attempt the in-chapter exercises and practice until you can do them fully on your own, just like in a test
3. Write down your questions and ask them in class

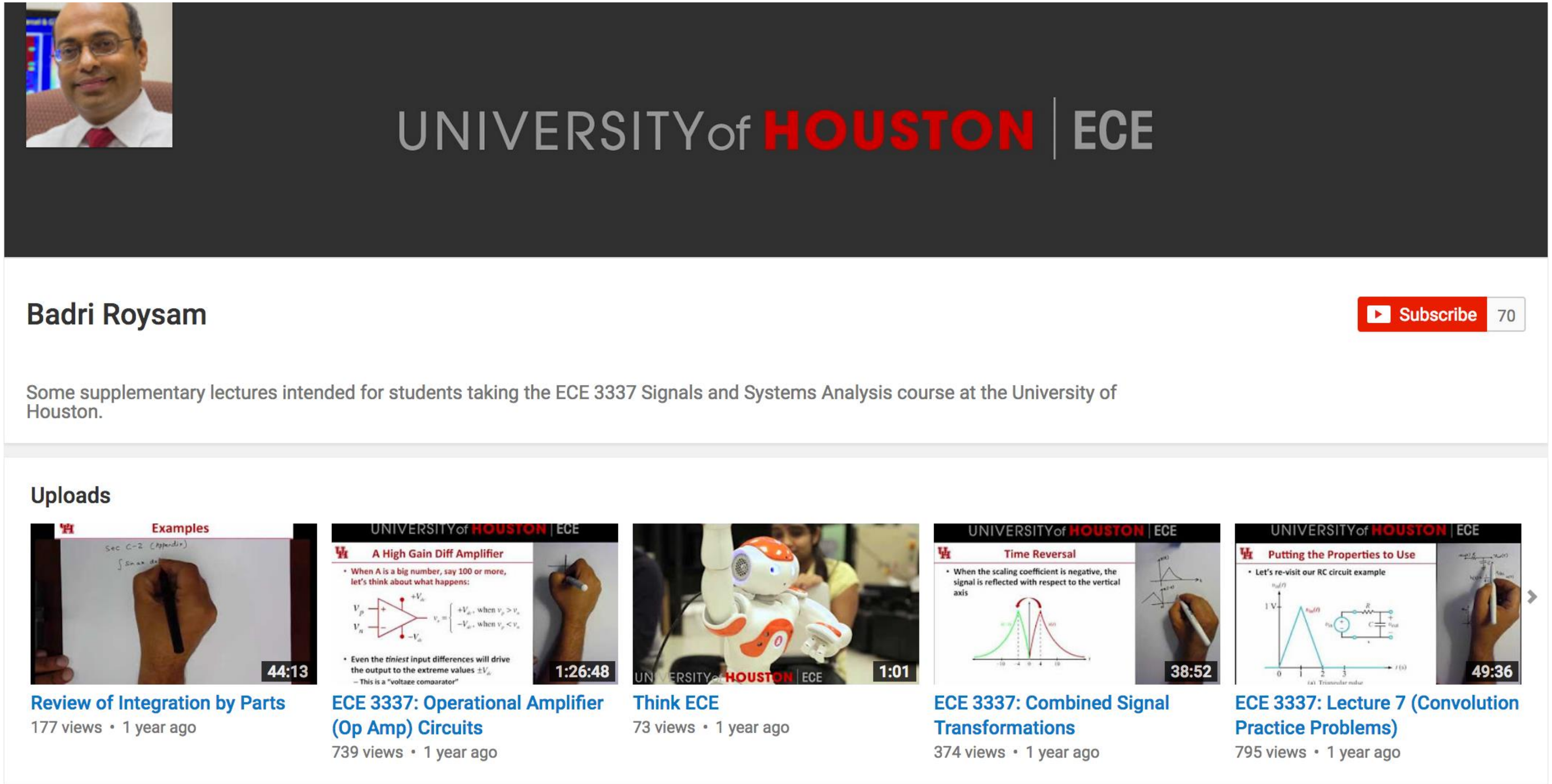
Classroom Rules

This course requires 100% attention and active engagement

- You must come prepared for each class as instructed
- Pencil and paper work best (pen tablet is ok)
- No cell phones, laptops, or other distracting gadgets in class <u>(offenders will be asked to leave)</u>
- Copy of lecture notes are provided to write on
- Do the problems when instructed
- Ask questions, get answers!

YouTube Channel

- We have recorded some video lectures on ECE 3337 related topics for you
- Search YouTube.com by "Badri Roysam"



Action Item #3: Work the Math Appendices

- Make sure that you can do all the examples in Appendix B on your own
- Make sure that you can derive all the formulas in Appendix C-2 and C-5 on your own
- Work through the background math review questions posted on Canvas, and make sure that you can get the correct answers for each question without looking at the solutions (pretend its an exam)

Microsoft Teams Site

We have a Microsoft Teams site for this class

- Remember to install and activate this App on your device(s)
- Provides extensive voice, video, and screen/file sharing facilities
- Use Teams to post questions privately or in group mode, speak with instructors/classmates, and have group discussions with your instructors/classmates
- We will use this tool to post recordings of lectures or help sessions
- We may use Teams as a backup for online sessions in case we have problems with Zoom

Essential Devices: Your Responsibility

You are required to have a working webcam, and sufficient network connectivity to support its use for online sessions

- We encourage you to have a document video camera or a tablet device with a stylus for interactive discussions

You are required to have a reliable PDF document scanner

- All worksheets need to be submitted on Canvas as PDF files
- You will need to scan your handwritten work, combine multi-page documents into a single PDF file, and upload the PDF file in a timely manner and reliably
- The scanned document should be checked before uploading to make sure that the grader can clearly read your answers

Please have backup devices and network connectivity options in place

Your Instructors

Classes	MW 2:30 – 4:00 pm
Room	Virtual initially, Room TBA
Instructor	Sebastian Csutak
Office	TBA
Phone	281-846-4028 Or via MS TEAMS
Email	scsutak@uh.edu
Office Hours	Wednesday 4:00pm – 6:30pm

Classes	MW 2:30 – 4:00 pm
Room	SEC 206
Instructor	Rohith Reddy
Office	Room W312 Engineering Building 2
Phone	713-743-7207 Or via MS TEAMS
Email	rkreddy@uh.edu
Office Hours	Mondays 4:00pm – 6:30pm

Both instructors are available to help you

We may run combined sections when needed

Grades

Item	Schedule	Points	Adjustment
Advance Preparation	One per lecture	10%	
Attendance	Random	5%	
Worksheets	One per lecture	15%	Lowest 2 dropped
Mid-term 1 (Ch. 1 – 2)	In Class	20%	
Mid-term 2 (Ch. 3 – 4)	In Class	20%	
Final Exam (Ch. 1 – 6)	Finals week, In Class	30%	

Advance preparation is on the Zybooks version of the textbook

Worksheets are Submitted Online on the UH Canvas System

- **Deadlines are firm, no exceptions**

Letter grade based on your overall score

Generous reward for effort

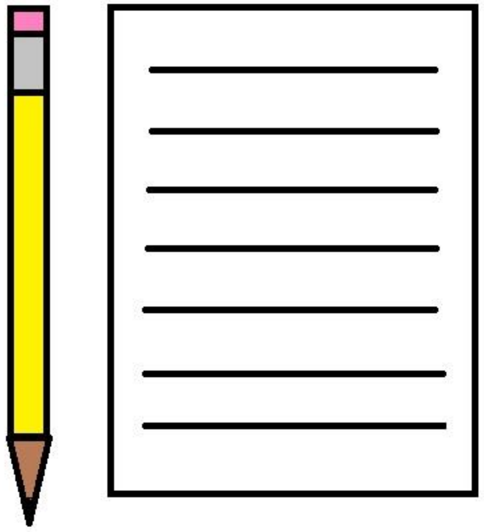
- **Attempt everything & do your best**

Be honest and stay out of trouble

- **Simply not worth taking chances!**

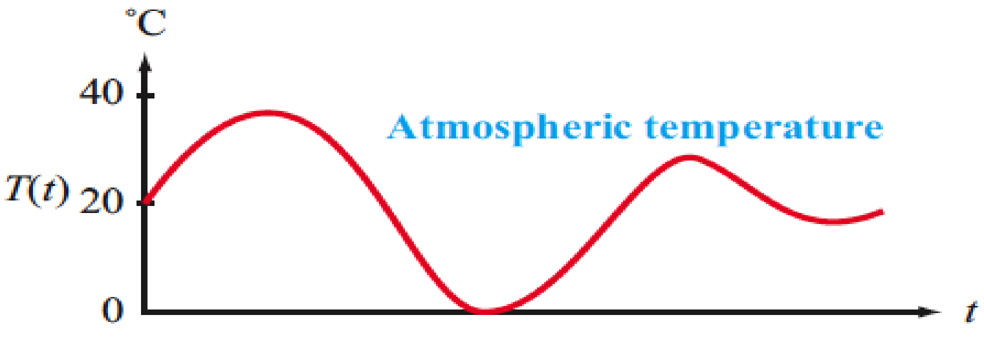
Course Policies

- This course is focused on learning fundamental concepts
  - Pencil and paper work best
  - Do the problems by hand *first*, then use calculators/software to verify
  - For the exams, calculators/computers/tablets are not allowed
- Please read the course handout carefully
  - First assignment: sign agreement
- Feel welcome to ask questions, get answers
- Give feedback – help me teach you better



Signal Representation

Representing a signal is the first step to signal processing



(a) Atmospheric temperature in °C

The temperature *T(t)* is plotted as it changes over time *t.*

We call this a ***waveform***

- This is an example of an **analog signal**, since the signal *T(t)* varies **continuously**, just like physical temperature
- This is also a **continuous-time signal**, since *t* varies continuously
- Such signals can theoretically capture every detail

Signal Transformations

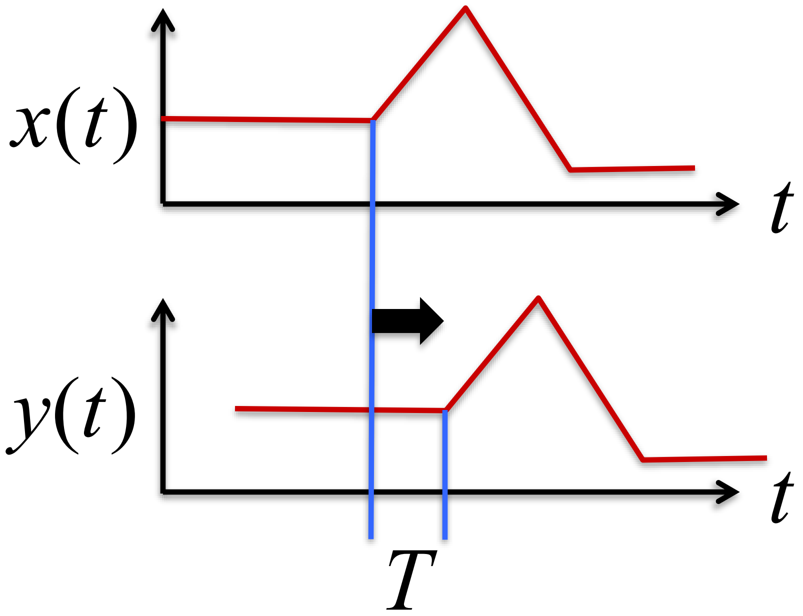
There are an infinite number of ways to transform signals

Input Signal → **Signal Transformation** → **New Signal**

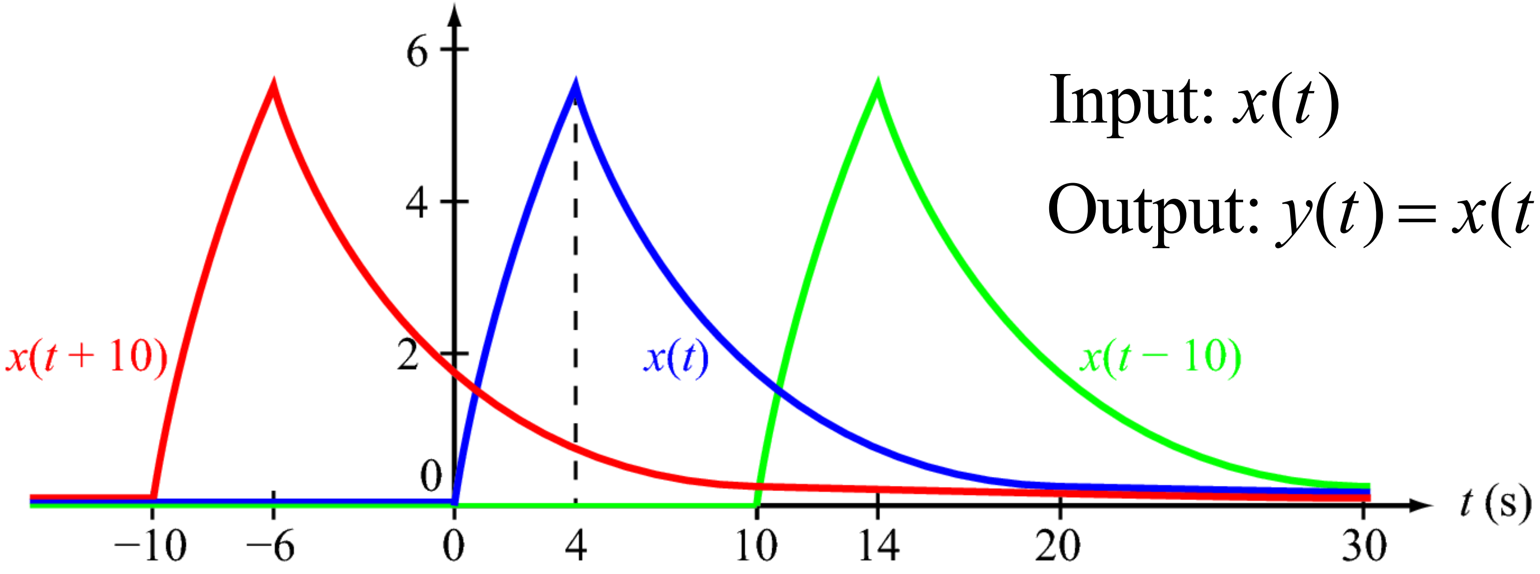
Example: Time shift a signal

Input: $x(t)$

Output: $y(t) = x(t - T)$



Example of Shift



Input: $x(t)$

Output: $y(t) = x(t - T)$

- The waveform is simply shifted along the *t* axis

$$T > 0 \text{ means shift to the right (delay)}$$

$$T < 0 \text{ means shift to the left (advance)}$$

- The **shape** of the signal is unchanged
- The **height (amplitude)** is unchanged

Example: Echo Effect

We can simulate an echo by adding a delayed version of a signal to itself

$$Input: x(t)$$

$$Output: y(t) = x(t) + \underline{0.5x(t-T)}$$

$$\text{Multiple Echoes: } y(t) = x(t) + \underline{0.5x(t-T) + 0.25x(t-2T)} + \ldots$$

“Hi There!”

$T = 1 \sec$

Example: Noise-canceling Headset

$s(t)$: The (clean) audio signal that we want to listen to

$n(t)$: The ambient noise picked up by the microphone

$\alpha\, n(t)$: The attenuated noise signal that reaches the inside of the ear cup

Δ: The time needed for sound to travel from the surface to the ear cup interior

$s'(t) = s(t) - \alpha\, n(t - \Delta)$: The modified signal that is fed to the speaker inside the ear cup

Time Scaling (playback speed)

$$y(t) = x(at)$$

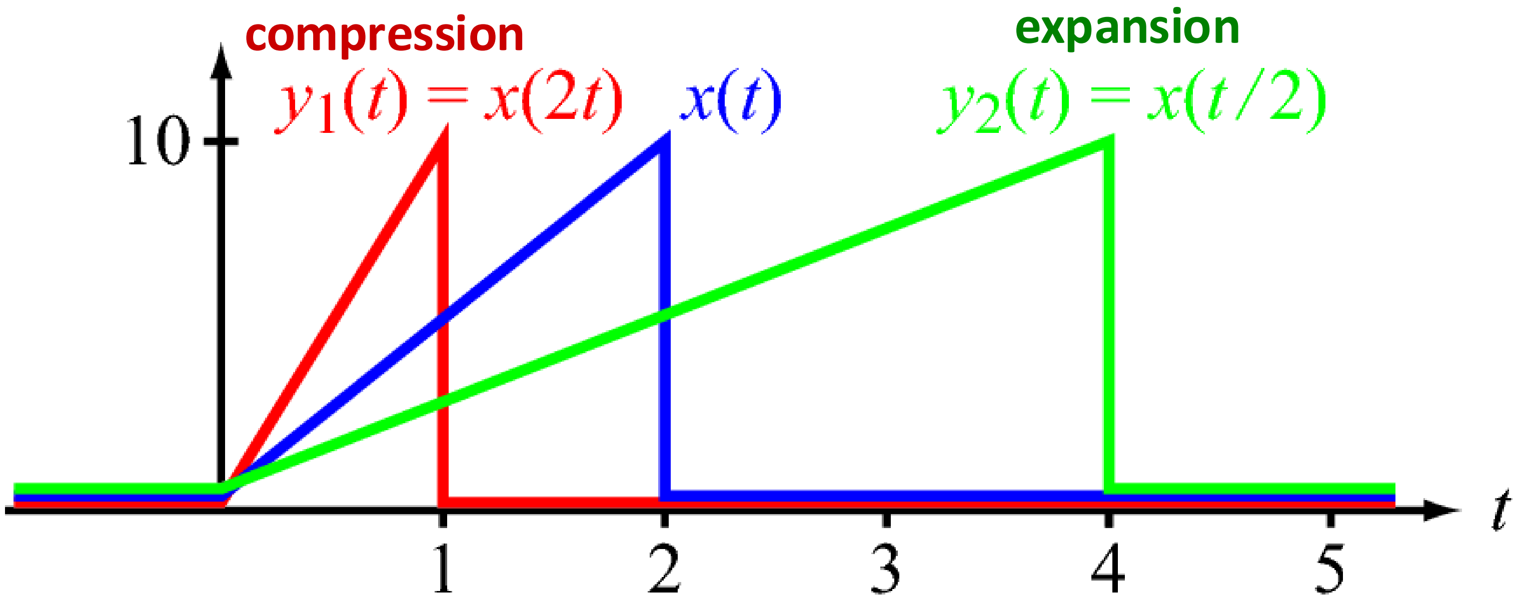
Scaling coefficient (a constant)

If $|a| < 1$, then $x(t)$ is expanded

If $|a| > 1$, then $x(t)$ is compressed

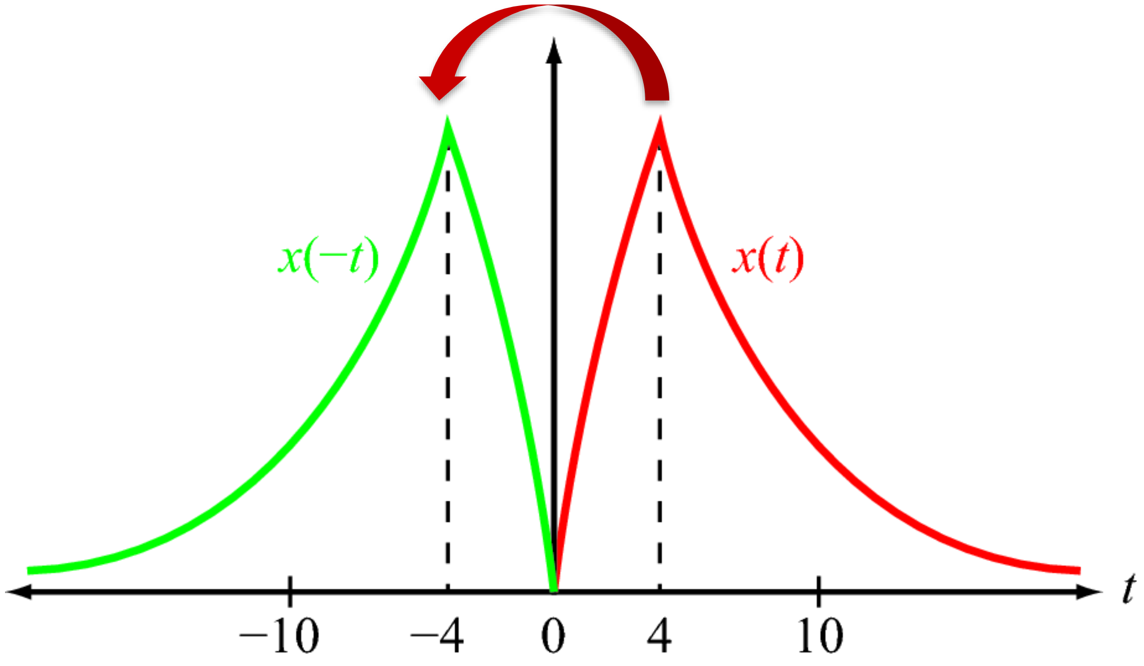
Example:

$$x(t) = \begin{cases} 5t & 0 \le t \le 2 \\ 0 & otherwise \end{cases}$$



Time Reversal

When the scaling coefficient is negative, the signal is played backwards



$$\text{If } a < 0, \text{ then } x(t) \text{ is reflected}$$

Time Scaling Summary

$$y(t) = x(at)$$

Scaling coefficient

(a is a constant)

- If $|a| < 1$, then $x(t)$ is expanded
- If $|a| > 1$, then $x(t)$ is compressed
- If $a < 0$, then $x(t)$ is also reflected

Think of *a* as "the playback speed"
At 2X speed the signal goes faster; at 1/2X speed, slower

Combined Transformation

Transformation recipes:

$$\text{Shift: } y(t) = x(t-T)$$ **Replace *t* by (*t*-*T*)**

$$\text{Scaling: } y(t) = x(at)$$ **Replace *t* by (*at*)**

$$\text{Reversal: } y(t) = x(-t)$$ **Replace *t* by (-*t*)**

We can do more complicated things by applying these transformations in sequence, for example consider:

$$y(t) = x(at-b)$$

Combined Transformation

$$y(t) = x(at - b)$$

Question: What sequence of the basic transformations will produce this output?

Approach #1:

$$x(t) \xrightarrow[\textbf{(Replace } \boldsymbol{t} \textbf{ by } \boldsymbol{at}\textbf{)}]{\textbf{Time Scale by } \boldsymbol{a}} x(at)$$

$$x(at) \xrightarrow[\textbf{(Replace } \boldsymbol{t} \textbf{ by (} \boldsymbol{t} \textbf{ - } \boldsymbol{T}\textbf{))}]{\textbf{Time Shift by T}} x(a(t - T))$$

$$a(t - T) = at - b \quad \square \quad T = \frac{b}{a}$$

Combined Transformation

$$y(t) = x(at - b)$$

Question: What sequence of the basic transformations will produce this output?

Approach #2:

$$x(t) \xrightarrow[\textbf{(Replace } t \textbf{ by } (t\,-\,b))]{\textbf{Time shift by } b} x(t-b)$$

$$x(t-b) \xrightarrow[\textbf{(Replace } t \textbf{ by } (at))]{\textbf{Time scale by } a} x(at-b)$$

BOTH APPROACHES ARE CORRECT

USE WHICHEVER IS MORE CONVENIENT

Important Reminders

Do only one transformation at a time by replacing t with its "adjusted version"

$$t \text{ replaced by } t-b$$

$$t \text{ replaced by } at$$

Don't mix them up in a single step

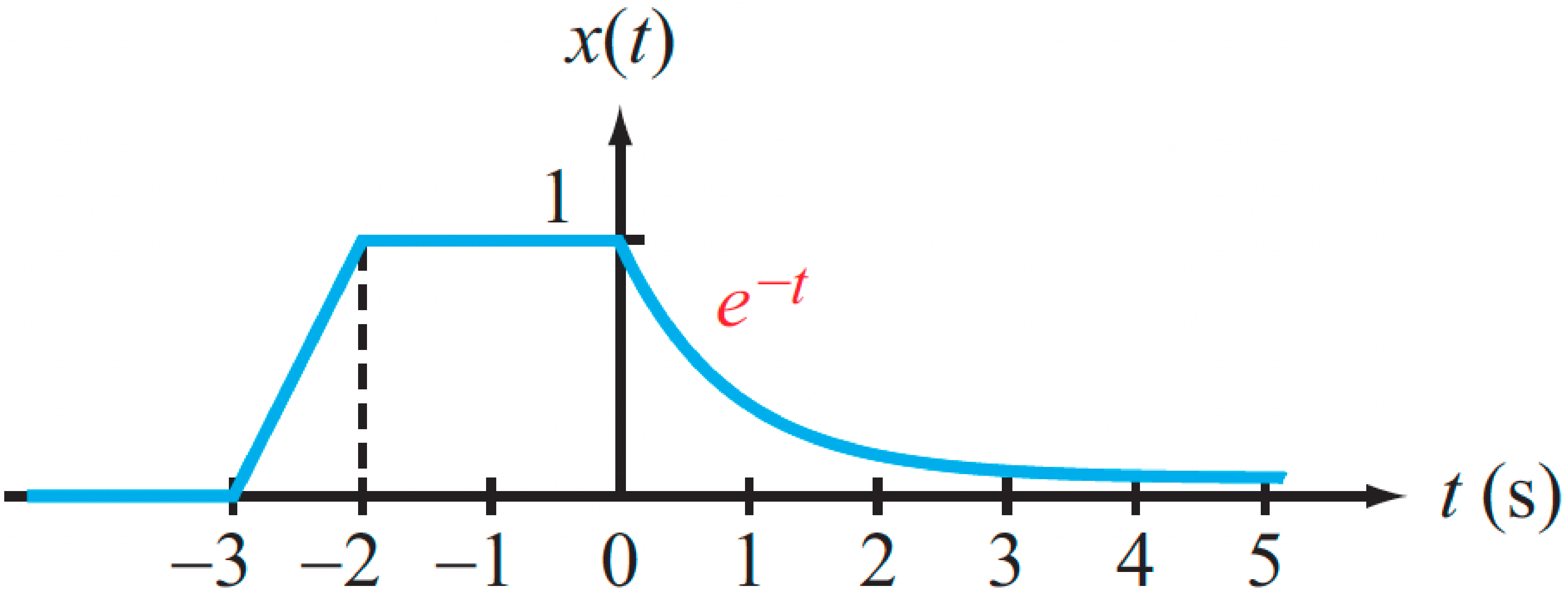
- **Can become confusing, and lead to mistakes**

Study the worked examples and practice

- **Question 1 of Exam 1 (old exams on Canvas site)**

Example

GIVEN:



PLOT:

$$y(t) = x(-2t + 6)$$

Approach #1

$$y(t) = x(-2t + 6)$$
$$= x(-2(t-3))$$

$x(t) \rightarrow$ **Time Scale by -2** $\rightarrow$ **Time Shift By 3** $\rightarrow y(t)$

System 1

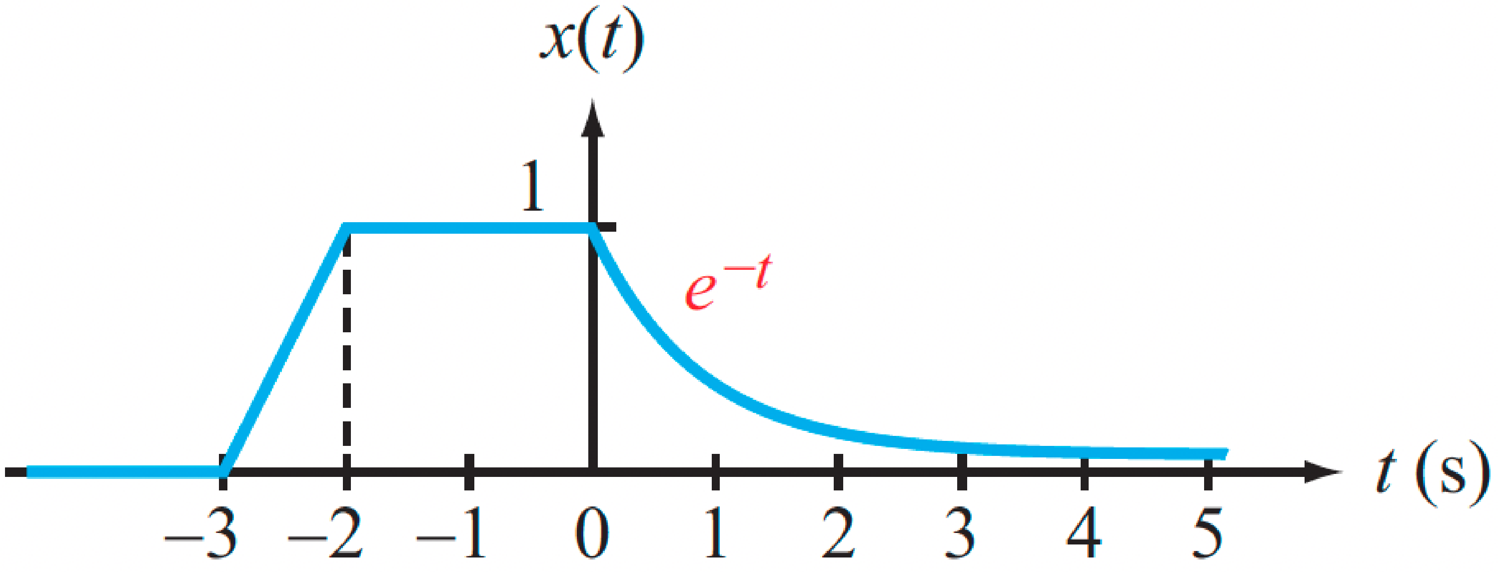
$$z(t) = x(-2t)$$

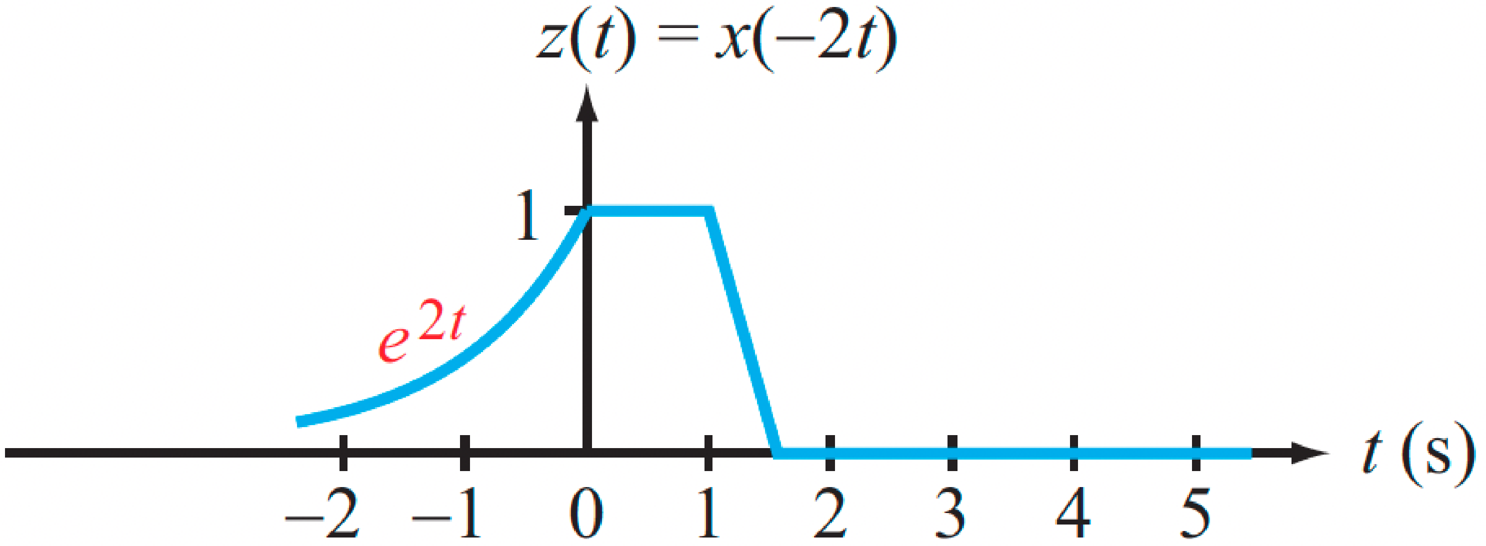
System 2

$$y(t) = z(t-3)$$

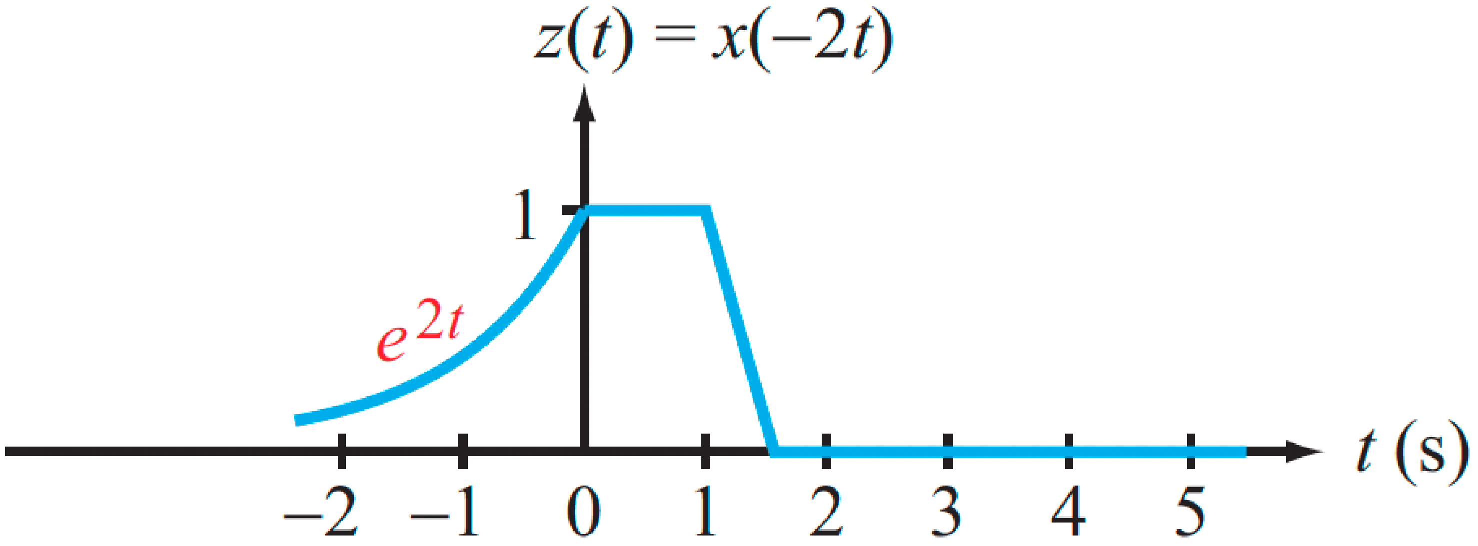
Time Scale by -2

GIVEN



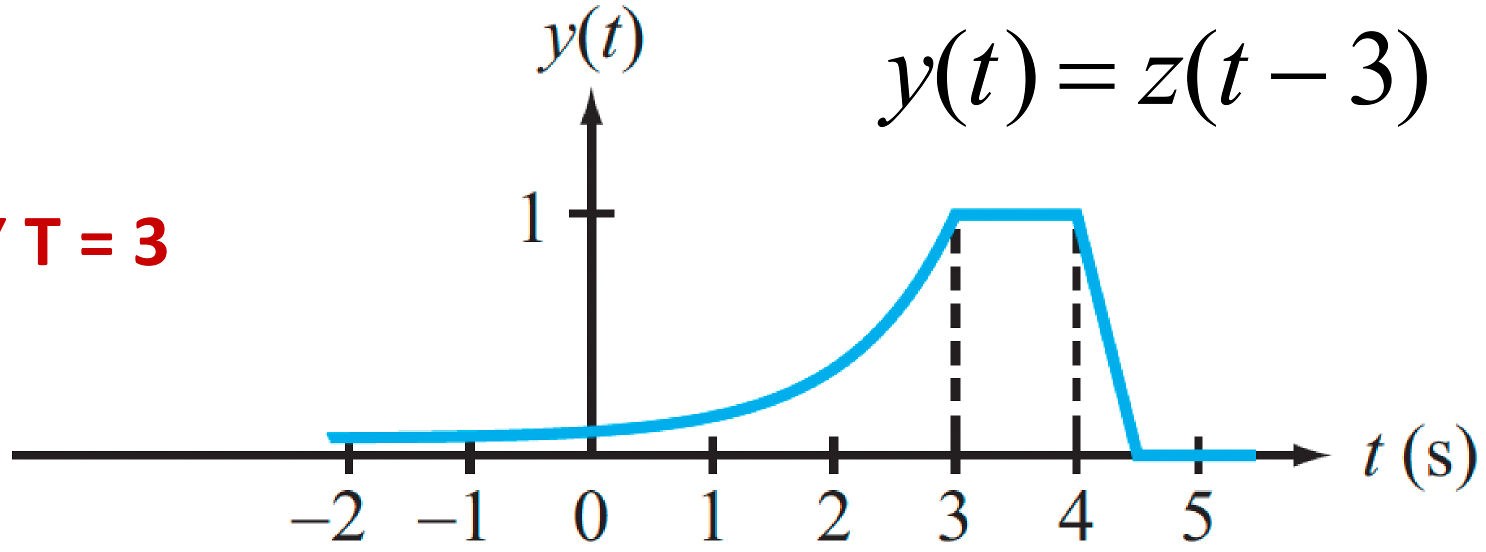


Shift by T = 3



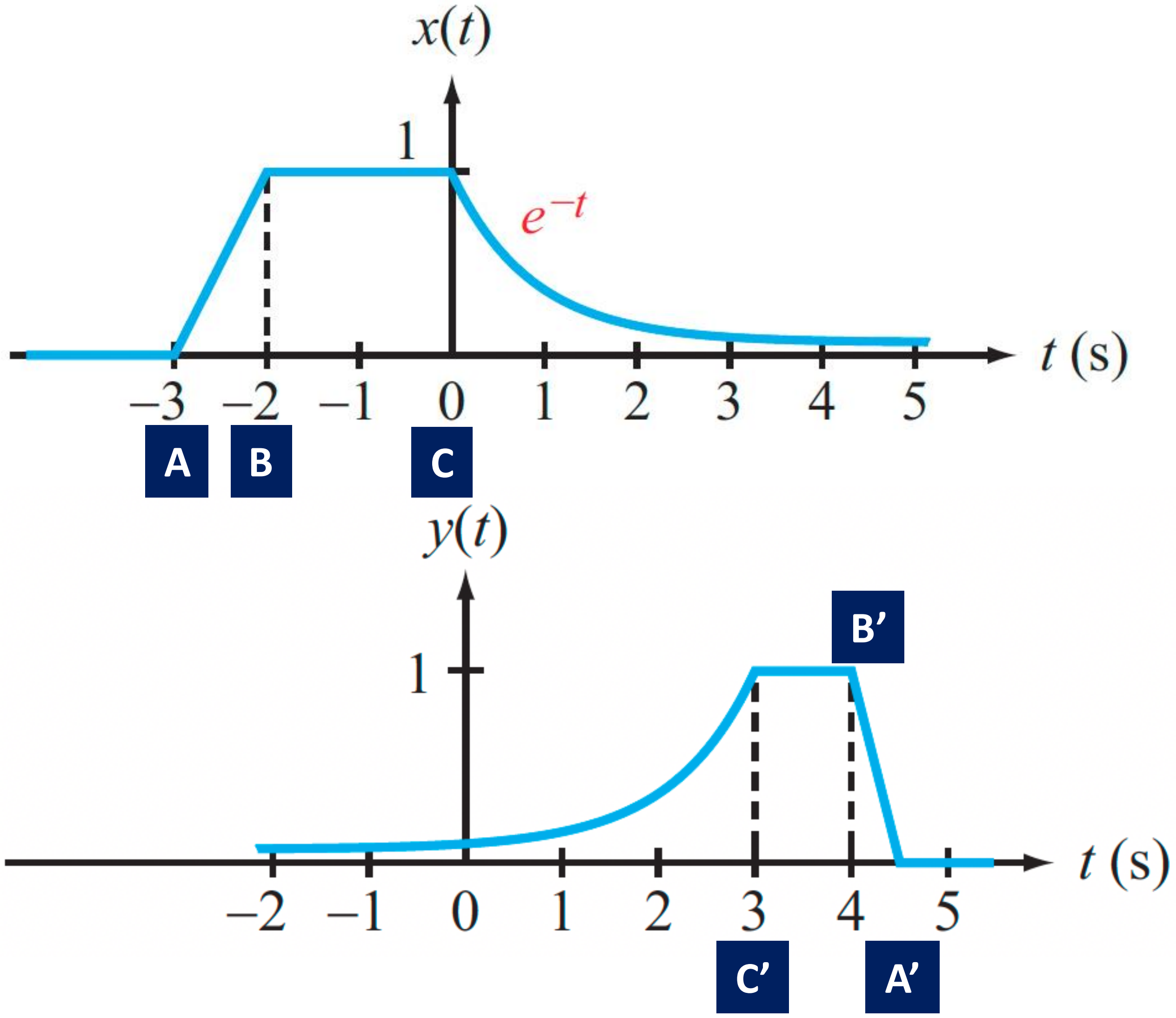
SHIFT BY T = 3

$$y(t) = z(t-3)$$



A Sanity Check Method

Check on identifiable "landmark points"



$$y(t) = x(-2t + 6)$$

$$t_A = -3$$

$$t_{A'} = 4.5$$

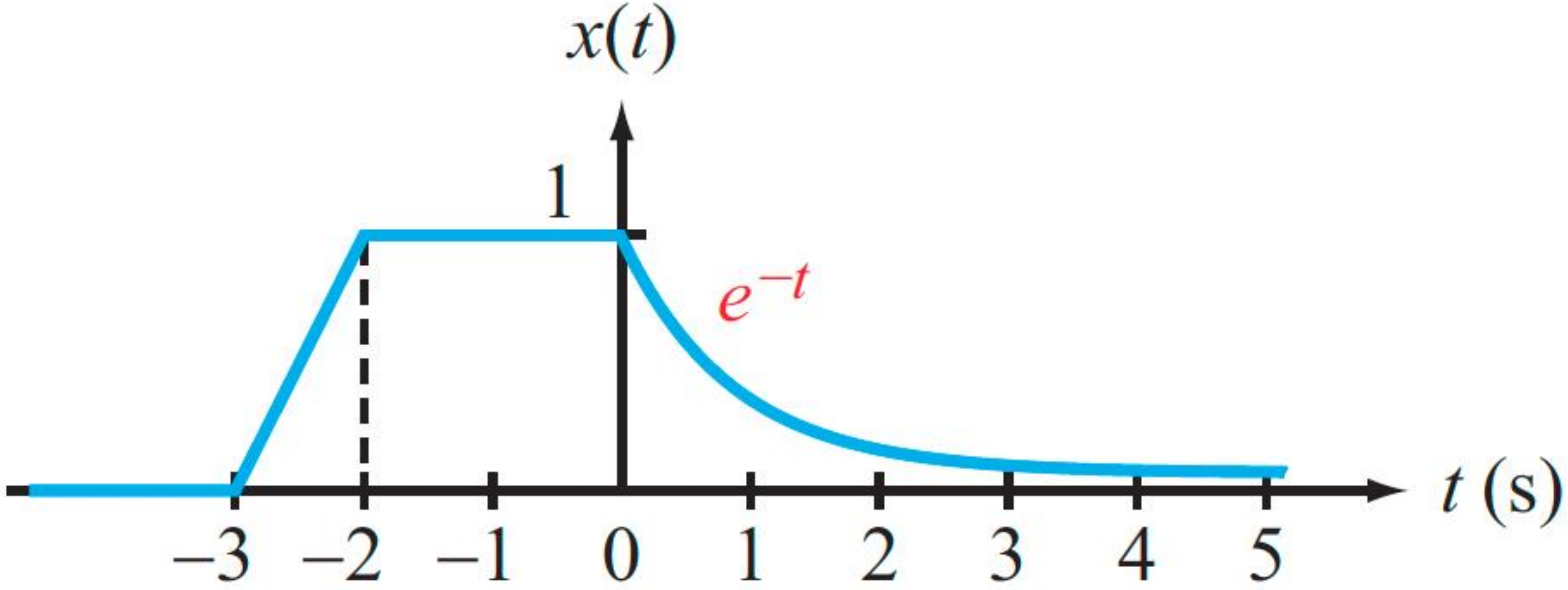
$$t_A = -2t_{A'} + 6$$

$$t_{A'} = \frac{(t_A - 6)}{(-2)}$$

In-Class Exercise

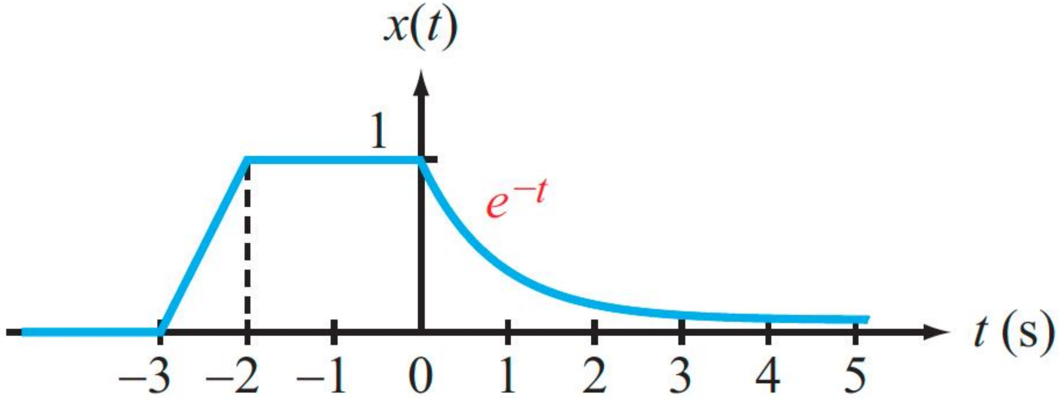
Show for the same example that shifting first and then time scaling yields the same answer

GIVEN:

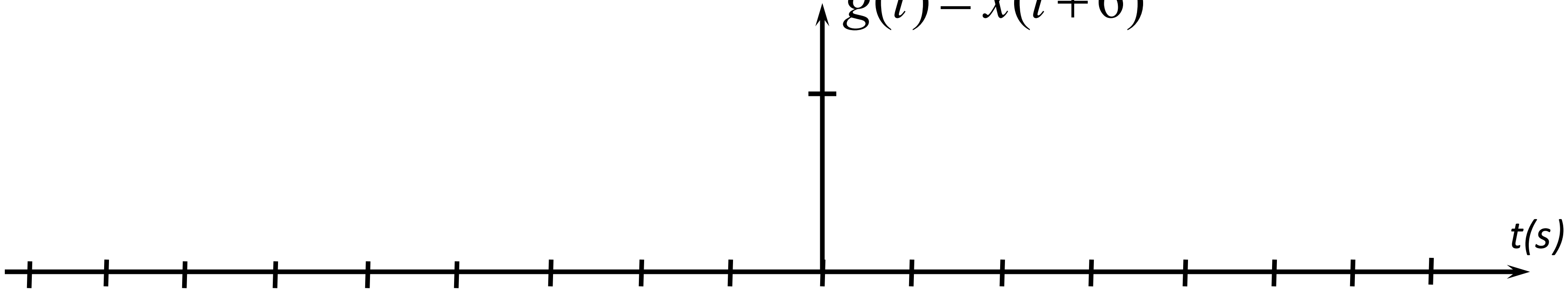


PLOT: $y(t) = x(-2t+6)$

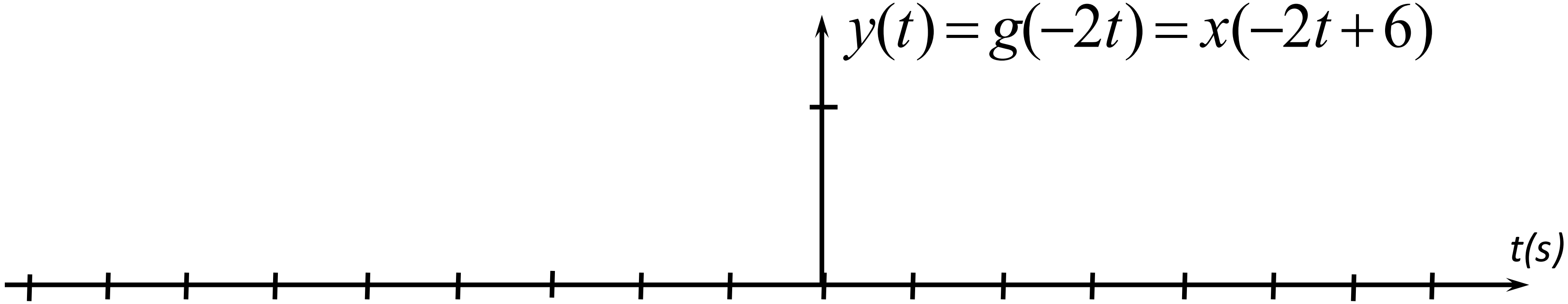
Workspace



$$g(t) = x(t+6)$$



$$y(t) = g(-2t) = x(-2t+6)$$



Concept Review Question

$$y(t) = x(-2t+6)$$

Which of the following transformation sequences are <u>incorrect</u>?
Two of the entries are done for you as examples

1. **Shift left by 6, then flip and compress by 2**
2. **Flip and compress by 2, then shift left by 3**

$$x(t) \rightarrow x(-2t) \rightarrow x\big(-2(t+3)\big) = x(-2t-6)$$

INCORRECT

3. **Shift right by 6, then flip and compress by 2**
4. **Shift left by 6, then flip and expand by 2**
5. **Flip and compress by 2, then shift right by 3**

$$x(t) \rightarrow x(-2t) \rightarrow x\big(-2(t-3)\big) = x(-2t+6)$$

CORRECT

Summary

This course is about signals and systems – they are everywhere

Input Signal → **System** → **Output Signal**

A system transforms a signal

We looked at basic signal transformations:

- Time Shift
- Time Scaling
- Time Reversal
- Combinations of the above (apply in sequence)

https://youtu.be/8X7YNnLiXjM

For the Next Class

Work through the textbook examples in Sections 1-1 – 1-2

Study Sections 1-3 – 1-4 ahead of Lecture 2 on Zybooks

How to Study for next class:

1. **Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts**
2. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
3. **Write down your questions and bring them to class**